# BT-102 (GS)

## B.Tech., I & II Semester Examination, June 2025

## Grading System (GS)

# MATHEMATICS-I

Time : Three Hours  Maximum Marks : 70

**Note :** (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8) — SET-A / Sem-2 subject

**1.** (a) Prove that a rectangular solid of maximum volume within a sphere is a cube. **7**

(b) If $u = \tan^{-1}[(x^3 + y^3)/(x - y)]$, then prove that $x\,\partial u/\partial x + y\,\partial u/\partial y$ = ... (source me RHS cut gaya tha). **7**

**2.** (a) Show that the surface area of the solid generated by revolution of the loop of curve $x = t^2$, $y = t - t^3/3$ about the x-axis is $3\pi$. **7**

(b) Change the order of integration in $\int\int dx\,dy / \log y$ and then evaluate it. **7**

**3.** (a) Find the Fourier Series for $f(x) = x + x^2$ in $(-\pi, \pi)$. **7**

(b) Find the half range sine series for $f(x) = x(\pi - x)$ in $(0, \pi)$. Hence deduce that $1/1^3 - 1/3^3 + 1/5^3 = \pi^3/32$. **7**

**4.** (a) Let $W_1$ and $W_2$ be subspaces of a vector space V with $W_1 \cap W_2 = \{0\}$. Let $w_1 \in W_1$, $w_2 \in W_2$ with $w_1 \neq 0$, $w_2 \neq 0$. Prove that $\{w_1, w_2\}$ is linearly independent. **7**

(b) Let $W_1$ and $W_2$ be subspaces of V. (i) Prove $W_1 \cap W_2$ is a subspace of V. (ii) Give an example to show $W_1 \cup W_2$ need not be a subspace. (iii) Is $W_1 \setminus W_2$ a subspace of V? **7**

**5.** (a) Find Eigen values and Eigen vectors of [[2, −2, 2], [1, 1, 1], [1, 3, −1]]. **7**

(b) Diagonalize the matrix [[3, −1, 1], [−1, 5, −1], [1, −1, 3]]. **7**

**6.** (a) Reduce the matrix [[3, 2, −1], [4, 2, 6], [7, 4, 5]] to the normal form, hence find its rank. **7**

(b) Find the first 3 terms in the Maclaurin series for (i) $\sin^2 x$ (ii) x e^(−x). **7**

**7.** (a) Investigate for what values of λ and μ the simultaneous equations x + y + z = 6, x + 2y + 3z = 10, x + 2y + λz = μ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions. **7**

(b) Find the Taylor series for the function $x^4 + x - 2$ centered at a = 1. **7**

**8.** (a) Evaluate $\int_0^\infty \int_x^\infty (e^{-y}/y)\,dy\,dx$. **7**

(b) If $x^x$ y^z = C then show that $\partial^2 z/\partial x \partial y = -(x \log e^x)^{-1}$. (Source text partially unclear.) **7**

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